



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination 2018
Higher 1

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H1 BIOLOGY

8876/01

Paper 1 Multiple Choice

20 September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid/tape.

Write your name, civics group and index number on the Multiple Choice Answer Sheet provided.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

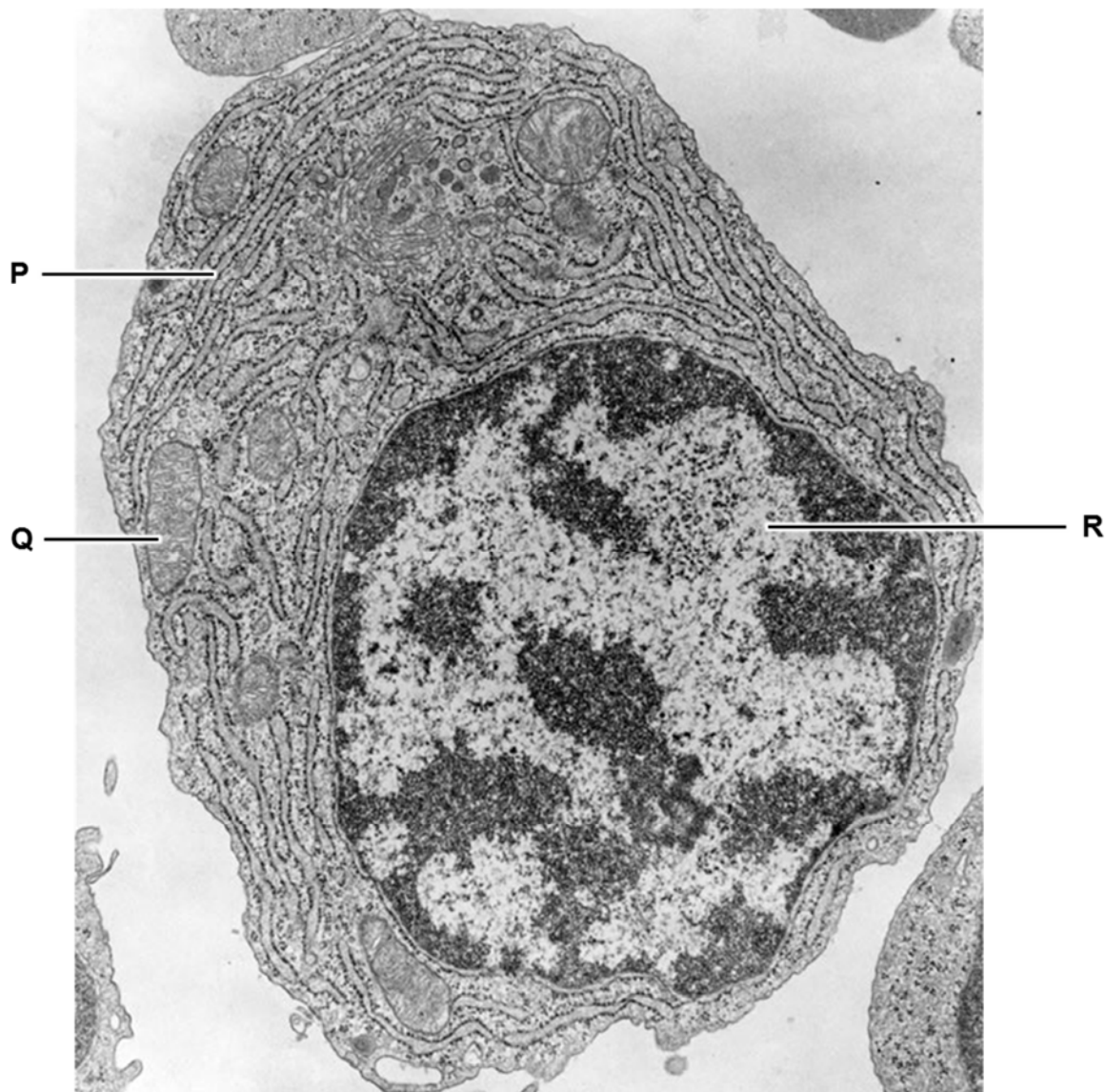
Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

At the end of the examination, you may bring home Paper 1 question paper.

QUESTION 1

The diagram shows an electron micrograph of a cell.



Which row correctly describes structures **P – R**?

	P	Q	R
A	synthesis of ribosomes	substrate level phosphorylation	active replication of genes
B	provision of large surface area for attachment of ribosomes	formation of ATP from light energy	active transcription of genes
C	synthesis of membrane proteins	oxidative decarboxylation	active transcription of genes
D	transport of proteins to Golgi apparatus	modification of mRNA transcripts	active replication of genes

QUESTION 2

A ribosome consists of a large and a small subunit, each subunit containing ribosomal RNA (rRNA) complexed with proteins.

Which sequence of events concerning ribosomes is correct?

- A** Within the nucleolus, rRNA and proteins are synthesised and subunits are formed. They become membrane bound as they are exported through the nuclear envelope to the cytoplasm and rough endoplasmic reticulum (rER).
- B** rRNA and proteins are synthesised in the Golgi body and are transported to the nucleolus for subunit formation.
- C** rRNA is synthesised in the nucleolus and proteins are synthesised by the rough endoplasmic reticulum (rER). Subunit formation occurs within the cytoplasm for free ribosomes and on the surface of the rER for attached ribosomes.
- D** rRNA synthesised within the nucleolus is complexed with proteins that have been imported from the cytoplasm. The subunits formed are exported to the cytoplasm via the nuclear pores.

QUESTION 3

Which two features contribute to the great tensile strength of cellulose?

- 1 glycosidic bonds linking the long chains of 1,4 α -glucose molecules
- 2 the –OH groups of the glucose molecules project outwards and form hydrogen bonds with neighbouring chains
- 3 the strength of the glycosidic bonds between the neighbouring chains of molecules
- 4 the successive glucose molecules are orientated at 180° to each other

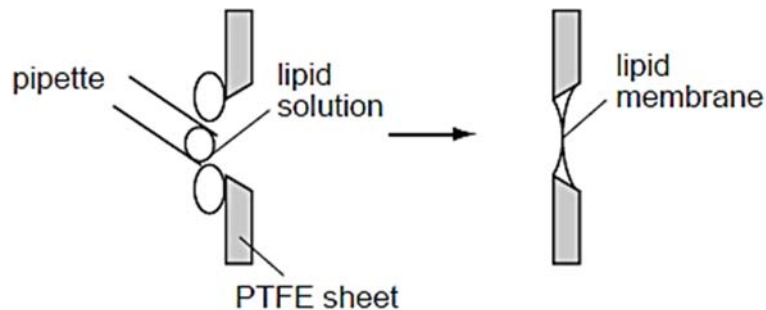
- A** 1 and 3 only **B** 1 and 4 only **C** 2 and 3 only **D** 2 and 4 only

QUESTION 4

Lipid membranes can be formed in the laboratory by painting phospholipids over a PTFE sheet with a hole in it.

Such a lipid membrane is impermeable to water-soluble materials including charged ions such as Na^+ or K^+ .

In one experiment with Na^+ ions, Na^+ ions did not flow across the membrane until a substance called gramicidin was added.



Which statement is consistent with this information and your knowledge of membrane structure?

Gramicidin becomes incorporated into the membrane and is

- A a carbohydrate molecule found only on the outside of the membrane.
- B a non-polar lipid which passes all the way through the membrane.
- C a protein molecule with both hydrophilic and hydrophobic regions.
- D a protein molecule which has only hydrophilic regions.

QUESTION 5

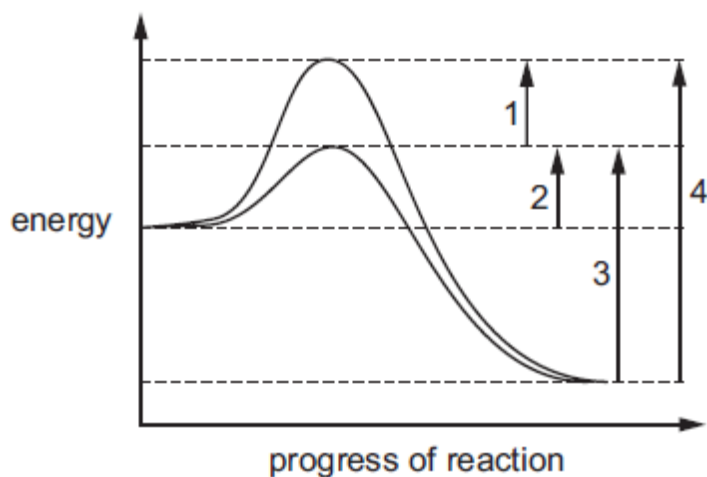
A symbiont may be defined as a species in which individuals live in a long-term, intimate and beneficial relationship with hosts of a different species. As the name suggests, endosymbionts live within their hosts.

Which statement provides the strongest evidence that mitochondria and chloroplasts in eukaryotes originated as prokaryotic endosymbionts?

- A Proteins encoded by the nucleus are exported to these organelles.
- B Their inner membrane has a different structure from other intracellular membranes.
- C They are surrounded by a double membrane.
- D They contain their own DNA and have 70S, rather than 80S, ribosomes.

QUESTION 6

The graph shows energy changes in a chemical reaction.

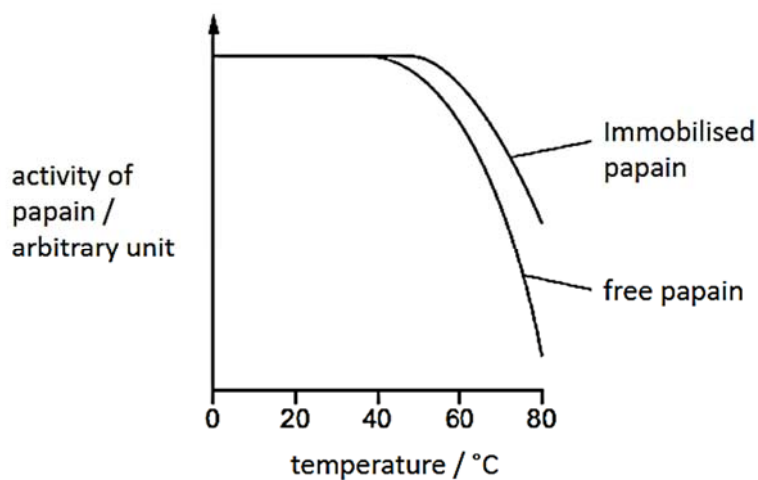


What is the activation energy when an enzyme is added?

- A** 1 + 2 **B** 2 only **C** 3 – 2 **D** 4

QUESTION 7

The graph compares the effect of temperature on the activity of the protease enzyme, papain, when in solution (free) and when immobilised in alginate beads.



Which statement about the effect of immobilisation of papain is correct?

- A** It increases the stability of papain at higher temperatures.
B It alters the shape of papain's active site at higher temperatures.
C It decreases the activity of papain at higher temperatures.
D It reduces the number of collisions of papain with the substrate.

QUESTION 8

The table contains results recorded by a student from an investigation into the effect of temperature on an enzyme-catalysed reaction. All other variables were standardised.

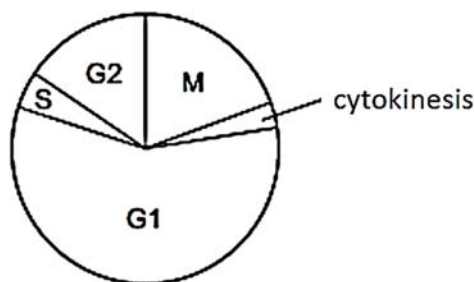
temperature / °C	rate of reaction / arbitrary units
10	3
20	7
30	16
40	33
50	32
60	14

What can be best concluded from the results?

- A The enzyme had the highest kinetic energy at 40°C.
- B The data for 50°C was anomalous.
- C The optimum temperature was between 30°C and 50°C.
- D The enzyme was held by only disulfide bonds at 60°C.

QUESTION 9

The diagram shows the cell cycle of a mammalian cell.



Checkpoints in the cell cycle of mammals prevent the cycle from continuing when mistakes are made or DNA is damaged.

Four of the checkpoints are described.

- 1 Mitosis is blocked if DNA replication is incomplete.
- 2 Anaphase is blocked if the assembly of chromatids on the spindle is unsuccessful.
- 3 DNA replication is blocked if DNA is damaged.
- 4 DNA replication stops if damage to DNA has not been repaired.

In which phases of the cell do these checkpoints occur?

	checkpoint			
	1	2	3	4
A	M	G1	S	G2
B	G2	M	G1	S
C	G2	S	G1	M
D	S	G2	M	G1

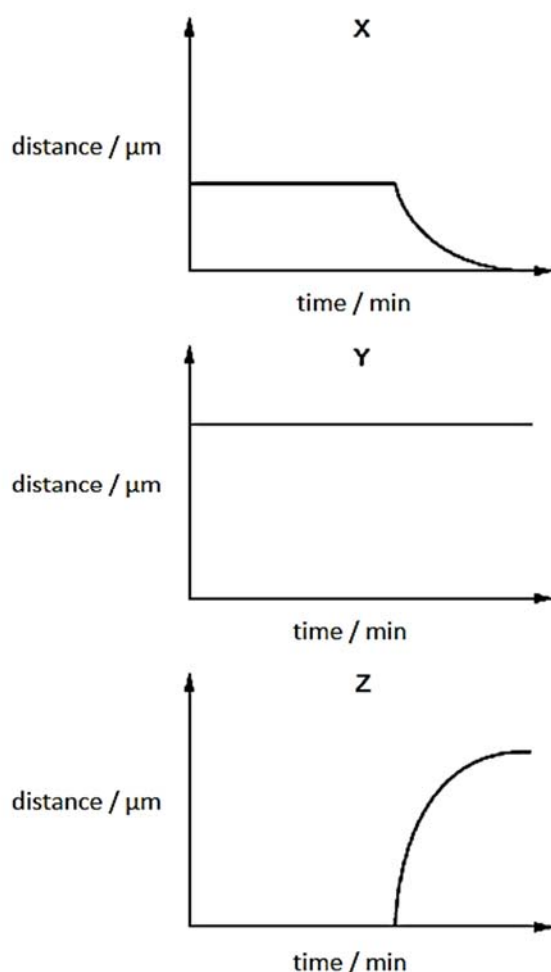
QUESTION 10

Which statement helps to explain how genetically identical cells are produced during the mitotic cell cycle?

- A** There is only one origin of replication, which ensures that DNA replication begins at a controlled site.
- B** Checkpoints of cell division are carefully regulated.
- C** Complementary base pairing occurs during DNA replication.
- D** RNA polymerase can correct some mistakes during transcription.

QUESTION 11

The graphs show various distance measurements taken from metaphase of mitosis onwards. The graphs are to scale when compared to one another.



Which row correctly identifies the distance measurement for each graph?

	X	Y	Z
A	distance between poles of spindle	distance between sister chromatids	distance of centromeres from poles of spindle
B	distance between poles of spindle	distance of centromeres from poles of spindle	distance between sister chromatids
C	distance of centromeres from poles of spindle	distance between poles of spindle	distance between sister chromatids
D	distance of centromeres from poles of spindle	distance between sister chromatids	distance between poles of spindle

QUESTION 12

Which molecules are found in both cytoplasm and the nucleus of a typical eukaryotic cell?

- 1 DNA nucleotides
- 2 DNA polymerase
- 3 RNA nucleotides
- 4 RNA polymerase

A 1, 2, 3 and 4 **B** 1, 2 and 4 only **C** 2, 3 and 4 only **D** 3 and 4 only

QUESTION 13

Two polynucleotide strands make up a DNA molecule.

What is a correct description?

- A** The percentage of cytosine is 50% of that of guanine in the whole molecule.
- B** The percentage of cytosine is the same as that of guanine in each strand.
- C** The percentage of cytosine is the same as that of guanine in the whole molecule.
- D** The percentage of cytosine is the same in each strand of the molecule.

QUESTION 14

Which of the following statements explain the difference in the direction in which the two strands of a DNA molecule are synthesised?

- 1 The replication of DNA is semi-conservative.
- 2 DNA polymerase can only add deoxyribonucleotides to the 3'OH group.
- 3 Each DNA molecule consists of two anti-parallel polynucleotides.
- 4 The synthesis of the lagging strand requires many more RNA primers.

A 1 and 2 only **B** 2 and 3 only **C** 1, 2 and 4 only **D** 2, 3 and 4 only

QUESTION 15

Scientists have made a nucleic acid (HNA) that has a sugar with the same number of carbon atoms as glucose instead of deoxyribose. Although genetic information can be stored by HNA, naturally occurring DNA polymerase cannot replicate HNA.

Which statements could explain why naturally occurring DNA polymerase cannot replicate HNA?

- 1 DNA polymerase cannot form bonds between the sugars of two HNA nucleotides.
- 2 DNA polymerase cannot form hydrogen bonds between two HNA nucleotides.
- 3 HNA nucleotides do not fit into the active site of naturally occurring DNA polymerase.
- 4 The shape of an HNA nucleotide is slightly larger than that of a DNA nucleotide.

A 3 and 4 only **B** 1 and 4 only **C** 2 and 3 only **D** 1, 2, 3 and 4

QUESTION 16

Some RNA molecules, called ribozymes, can catalyse reactions in a similar way to protein enzymes.

Most of these ribozymes have other RNA molecules as their substrates and catalyse reactions that break specific sugar phosphate bonds in the substrate molecules.

Which statements about these ribozymes are correct?

- 1 Hydrogen, ionic and disulfide bonds will be involved in the ribozyme structure.
- 2 The active site of a ribozyme is formed from a specific sequence of nucleotides.
- 3 Ribozymes can be formed because RNA can have a specific three-dimensional conformation.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

QUESTION 17

The codons UGU and UGC code for the amino acid cysteine, which can form disulfide bonds in a polypeptide.

The codon UGG codes for the amino acid tryptophan, which does not contain a sulfur atom.

The codon UGA is a stop signal.

The DNA triplet code for the 10th amino acid in a particular polypeptide is ACA.

Which single base substitutions in this triplet code will result in **no** disulfide bond being formed with the 10th amino acid in the polypeptide?

- A** ACC and ACG
B ACG and ACT
C ACT and ACC
D ACT only

QUESTION 18

Exceptions to the universal genetic code are found in mammalian mitochondria, as shown in the table.

mRNA codon	in mammalian cytoplasm, codes for	in mammalian mitochondria, codes for
AGA	arginine	stop
AGG	arginine	stop
AUA	isoleucine	methionine
UGA	stop	tryptophan

A short length of DNA triplet code with the following base sequence was transcribed.

TATTCTTCCACT

How many peptide bonds would be formed by ribosomes during translation in mammalian cell cytoplasm and in mammalian mitochondria?

	mammalian cell cytoplasm	mammalian mitochondria
A	2	1
B	2	0
C	3	0
D	3	1

QUESTION 19

Which statement(s) describe(s) how a gene mutation can lead to the production of a non-functional protein?

- 1 During transcription an incorrect nucleotide is added to a DNA molecule.
- 2 A codon in the mRNA transcribed from the mutated gene is changed.
- 3 The order of the bases in an anticodon on tRNA is altered during translation.
- 4 The sequence of nucleotides in the promoter of the gene is altered.

A 2 only **B** 1 and 2 only **C** 2 and 4 only **D** 2, 3 and 4 only

QUESTION 20

The following discoveries were made about myostatin, a protein that is produced in mammalian skeletal muscle cells.

- Myostatin circulates in the blood and acts on muscle tissue to slow down further differentiation and growth of muscle cell precursors called myoblasts.
- In racehorses that were bred true, a mutation involving the substitution of a single nucleotide has been identified in the *MSTN* gene which codes for myostatin.
- At the site of this mutation, the DNA nucleotide has either a cytosine (C) base or a thymine (T) base, giving race horses three possible genotypes for this mutation: CC, CT or TT.
- At two years of age, racehorses with the *MSTN* CC genotype have greater muscle mass than those with the TT genotype.

Which of the following statements may be best concluded from the above?

- A** Racehorses with the CC genotype contain hyperactive myostatin protein in their muscle tissue.
- B** In racehorses with the CC genotype, there is a change in the reading frame of the mRNA codon from the site of the mutation.
- C** All racehorses with the CC genotype are able to run faster than those with the CT and TT genotype.
- D** Racehorses with the TT genotype develop more muscles later in life.

QUESTION 21

The Himalayan variety of rabbits has white hair on the body and black hair on the feet, tail, ears and face. The allele for the Himalayan rabbit pigment pattern, c^h , is recessive to the alleles for normal colour (all hair agouti), C, as well as dark chinchilla (all dark grey hair), c^{chd} , and is dominant to the allele for albino (all white hair, no pigment production), c. All of the alleles of this gene produce different versions of the same enzyme involved in pigment production.

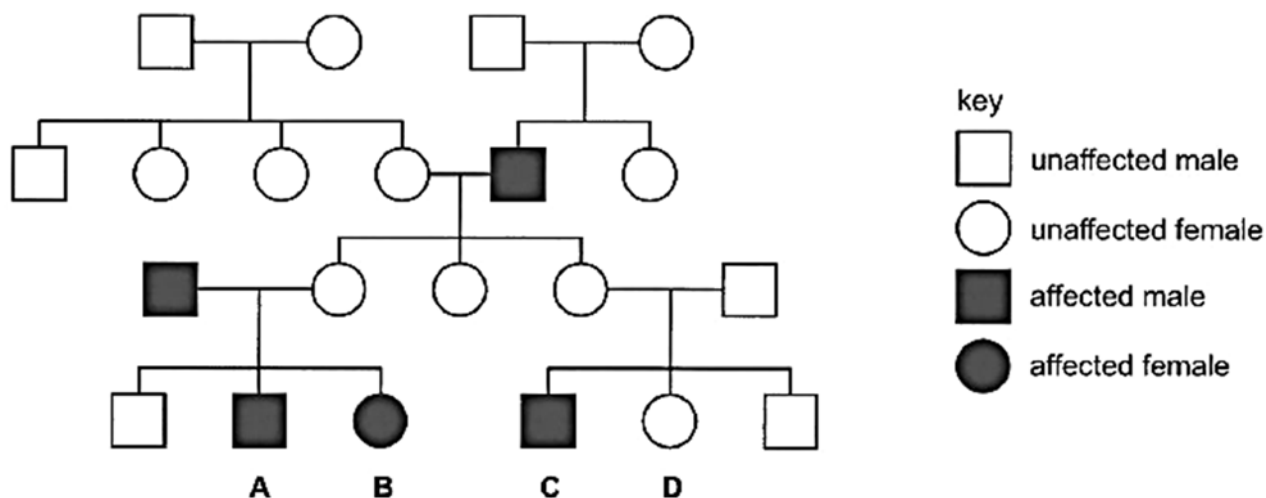
A patch of white fur was removed from a Himalayan rabbit and an ice pack secured to the skin. The fur that grew back on the patch was black.

Which of the following statements is correct?

	Genotypes of Himalayan rabbits	Explanation for pigment pattern in Himalayan rabbits
A	$c^h c^h$ only	The enzyme is denatured at the high skin temperatures found on the rabbits' bodies
B	cc only	The enzyme becomes inactive at the low skin temperatures found on the rabbits' feet, tail, ears and face
C	$c^h c^h$ and $c^h c$ only	The enzyme is denatured at the high skin temperatures found on the rabbits' bodies
D	$c^h c^h$ and $c^h c$ only	The enzyme becomes inactive at the low skin temperatures found on the rabbits' feet, tail, ears and face

QUESTION 22

The pedigree diagram shows the inheritance of an X-linked recessive trait in humans. If the recessive allele is r , which individual possesses the genotype rr ?



QUESTION 23

Uncontrolled cell division can form tumours.

Which statement is correct for tumour cells **only**?

- A Metaphase does not take place.
- B Cytokinesis does not occur.
- C Interphase takes less time.
- D They have mutated DNA.

QUESTION 24

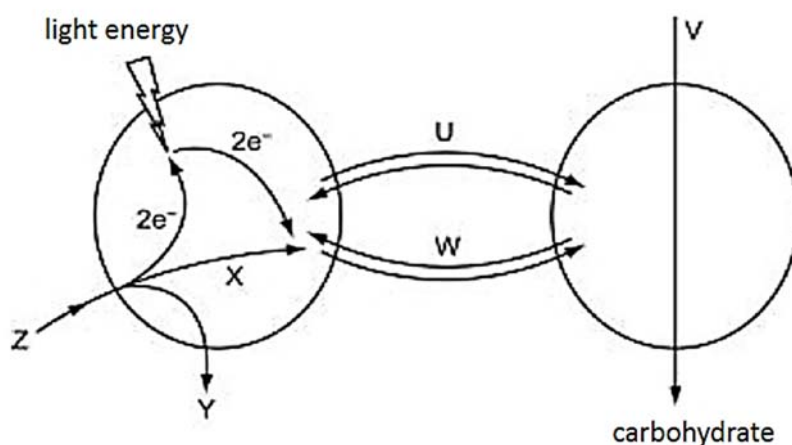
What are the products of glycolysis?

- 1 pyruvate
- 2 reduced FAD
- 3 reduced NAD
- 4 reduced NADP

- A 1 and 2 B 1 and 3 C 2 and 3 D 2 and 4

QUESTION 25

The diagram summarises the reactions of photosynthesis in a plant.



Which of the following correctly identifies the substances involved?

	CO_2	reduced NADP	H_2O	ADP	2H	O_2
A	U	W	Y	V	Z	X
B	U	X	Z	W	Y	Z
C	V	U	Z	W	X	Y
D	V	W	Y	U	X	Z

QUESTION 26

Some apples can be stored in controlled atmospheric conditions for up to a year. Taste and texture are maintained by using conditions that reduce the production of a fruit-ripening plant hormone while limiting the build-up of ethanol. Ethanol damages the fruit.

The storage conditions needed include low temperature (1°C), high carbon dioxide concentration (1.2%) and low oxygen concentration (0.9%).

Why are these conditions needed?

- 1 Low oxygen concentration favours anaerobic respiration.
- 2 Enzyme activity is reduced.
- 3 Conversion of sugar to ethanol is minimised.
- 4 High carbon dioxide concentration promotes photosynthesis.

A 1, 2 and 3 **B** 1, 2 and 4 **C** 2 and 3 only **D** 3 and 4 only

QUESTION 27

Which of the following describe processes that lead to an increase in variation?

- 1 breaking and re-joining in homologous chromosomes during prophase I of meiosis
- 2 random distribution of homologous chromosomes to the cell poles during anaphase I of meiosis
- 3 random variation in allele frequency with time that may result in alleles becoming more common
- 4 the production of new alleles by substitution of one base for another in DNA

A 1 and 4 only **B** 2 and 3 only **C** 1, 2 and 4 only **D** 1, 3 and 4 only

QUESTION 28

The skin of a small amphibian contains a substance, tetrodotoxin (TTX), that is deadly to all predators except the common garter snake. TTX acts by blocking membrane ion channels involved in muscular movement. Some of the garter snakes slow down for a few hours after eating the amphibian, but are not otherwise harmed.

Some variants of the amphibian have higher toxicity because of higher TTX production, while some variants of the garter snake have higher resistance. Where snakes have not been exposed to TTX, the snake variants with higher resistance have been observed to move more slowly than those with normal resistance.

From the information provided above, what is **not** a valid suggestion concerning natural selection and evolution in the amphibian and the garter snake?

- A** A selection pressure acting against the garter snake with higher resistance is the amphibian variant with normal toxicity.
- B** Locations with the most toxic amphibian variants are likely to have the most resistant garter snake variants.
- C** Predation by garter snakes is the selection pressure for increasing toxicity of the prey; the selection pressure acting against the most resistant garter snakes is predation.
- D** TTX resistance is likely to involve a mutation in the gene coding for the membrane ion channels.

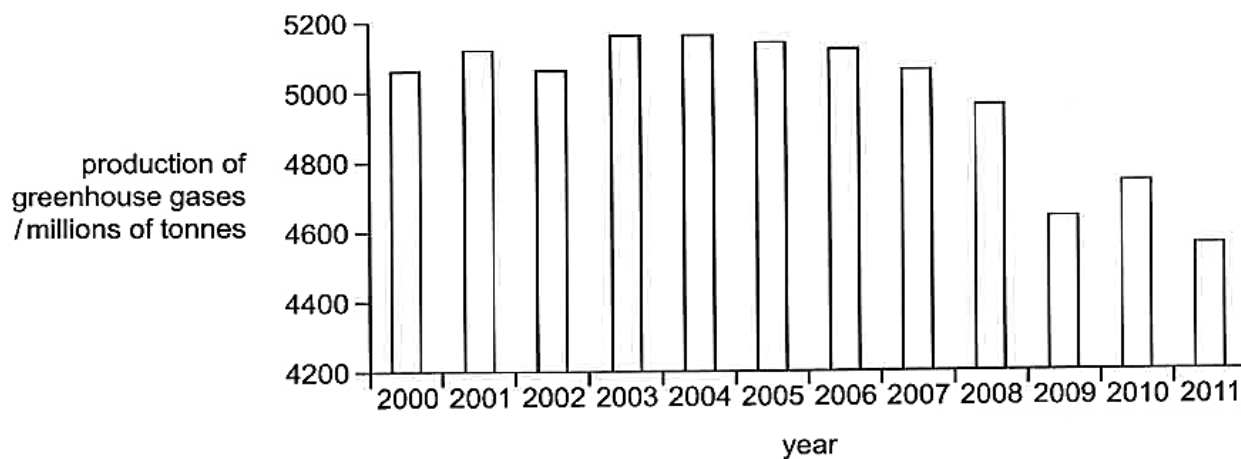
QUESTION 29

By which mechanism do greenhouse gases contribute to global warming?

- A** Their higher concentration absorbs more long wave radiation coming from the sun.
- B** Short wave radiation emitted from the Earth's surface increases with their concentration.
- C** They absorb higher amounts of long wave radiation emitted from the Earth's surface as their concentration increases.
- D** They absorb higher amounts of short wave radiation caused by increased combustion of fossilized organic matter.

QUESTION 30

The bar chart shows the production of greenhouse gases (carbon dioxide and methane) from agriculture in the European Union (EU) from 2000 to 2011, measured in millions of tonnes.



Which of the following could contribute to the trend seen between 2003 and 2009?

- A conversion of intensive farmland into woodland reserves
- B greater use of agricultural machinery for harvesting
- C increased consumption of meat-based products
- D increased import and export of crops between EU countries

End of Paper

H1 Biology Prelim Paper 1 Answers

1. C	6. B	11. C	16. D	21. C	26. C
2. D	7. A	12. A	17. C	22. B	27. C
3. D	8. C	13. C	18. B	23. C	28. A
4. C	9. B	14. B	19. A	24. B	29. C
5. D	10. C	15. A	20. D	25. C	30. A